HD Map Localization

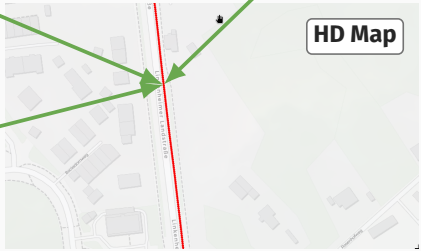
Michael Schweitzer

June 14, 2026

HD Map Localization



sensor data
EGO
IMU
Stereo
...



IBM Rational DOORS — Formal module: 'HD Map Requirements' — /HD Map/Localization

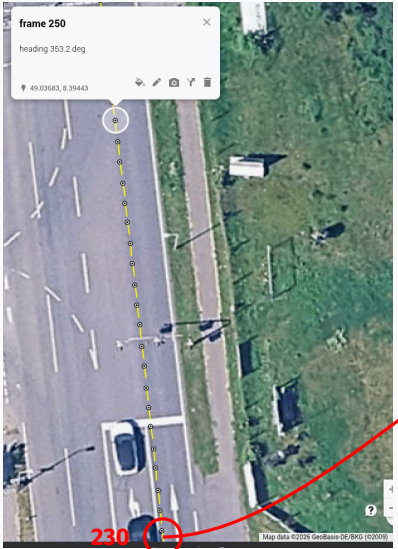
File Edit View Insert Link Analysis Table Tools Discussions Change User Help

View: Standard view

#	ID	Object Text
2		**2 Localization requirements**
2.1	**HD-001**	The map-relative accuracy error shall not exceed 0.1 m.
2.2	HD-002	The localization estimate shall be available at 10 Hz.
2.3	HD-003	The localization latency shall not exceed 50 ms.

Username: mschweitzer — Exclusive edit mode — 1 object selected

HD Map Localization - GPS is not the Solution!



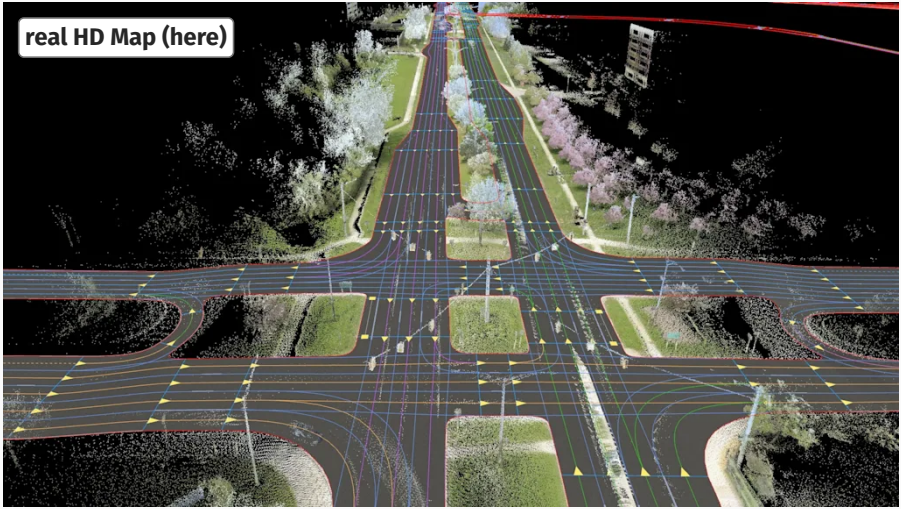


KITTI used a combined GPS/IMU system with **Real-Time Kinematic (RTK) correction**, which provides a localization accuracy/error of approximately **5 cm.** KIT - Karlsruher Institut für Technologie +1

Die Kontinentaldrift in Europa beträgt für den Zeitraum 2011–2026 (15 Jahre) insgesamt etwa **15 cm bis 60 cm**, da sich die eurasische Landmasse je nach Region um **1 cm bis 4 cm pro Jahr** verschiebt. Geo.de +1

Die GPS-Genauigkeit von Smartphones auf Google Maps liegt im Freien meist bei etwa **5 bis 20 Metern** Google Help. Satellitenbilder bieten in Städten teilweise eine Auflösung von bis zu **50 cm**, die absolute Positionsgenauigkeit der Karten kann durch Verzerrungen aber um **bis zu 10 Meter oder mehr** abweichen. Google Help +2

HD Map Localization



Outline

- Introduction ✓ *(already done!)*
- HD Map Building *(prototyped)*
- HD Map Localization *(prototyped)*
- Thoughts
- Questions
- Links
  - github code
  - github slides

Introduction

Idea

- One reason
- Another reason
- A third one with emphasis

Setting

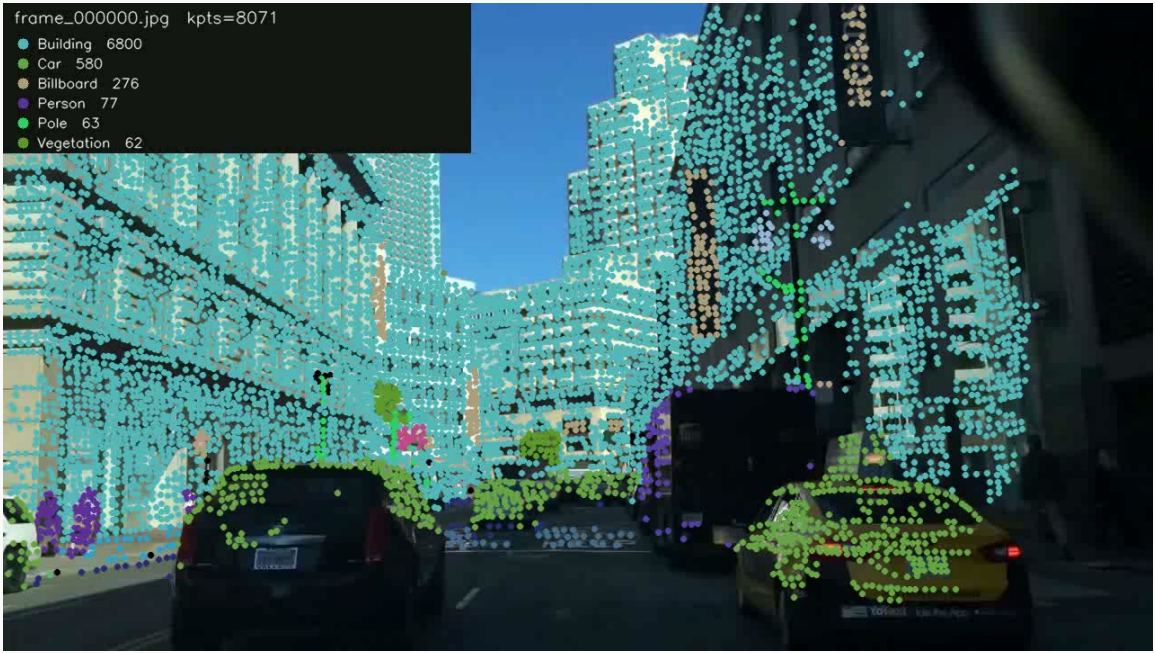
- One reason
- Another reason
- A third one with emphasis

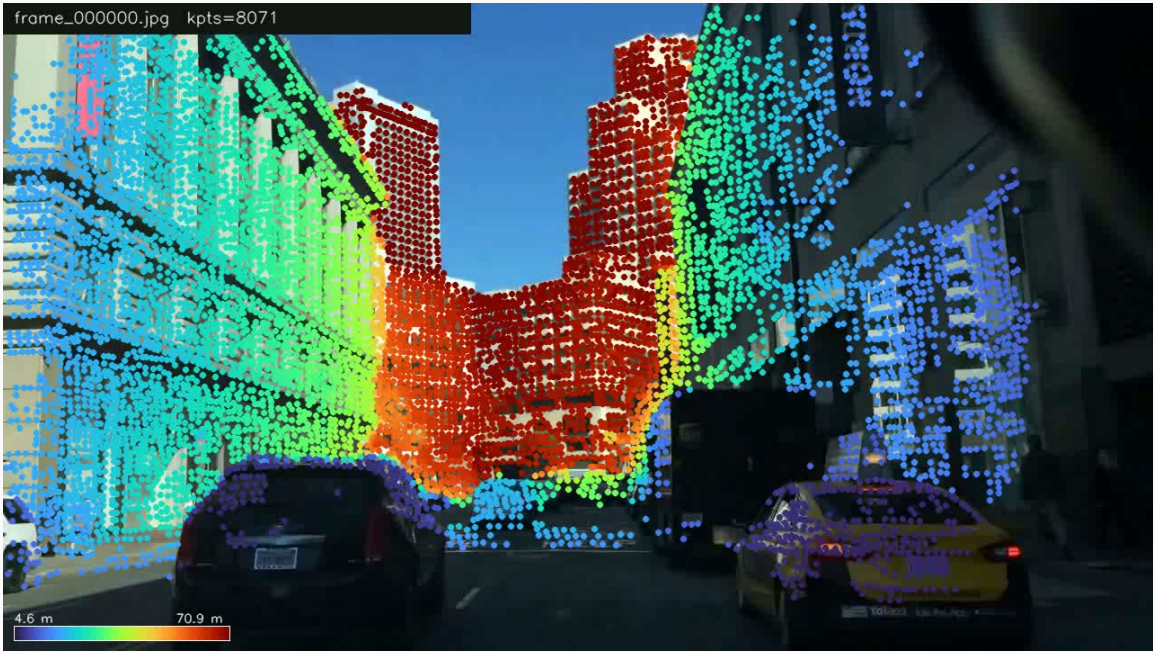
Approach

- One reason
- Another reason
- A third one with emphasis

A video

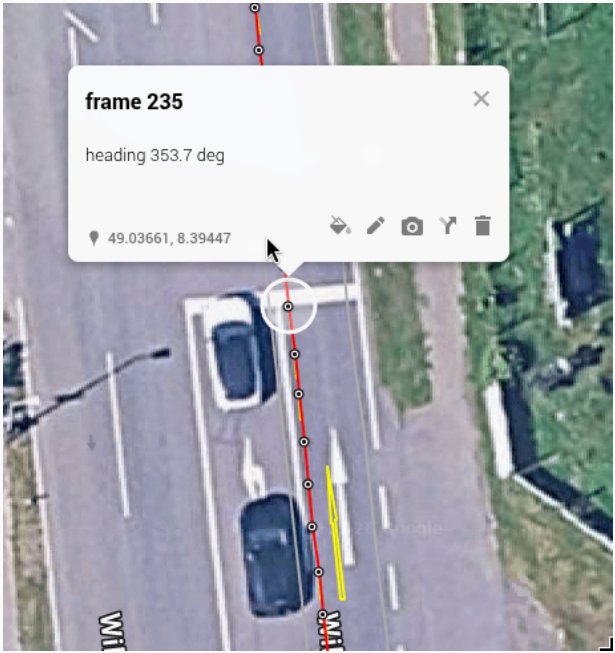

perception cache v1 step=3
frame 000000 / 000099 npz: yes
semantic
instance
mono_depth
depth_conf

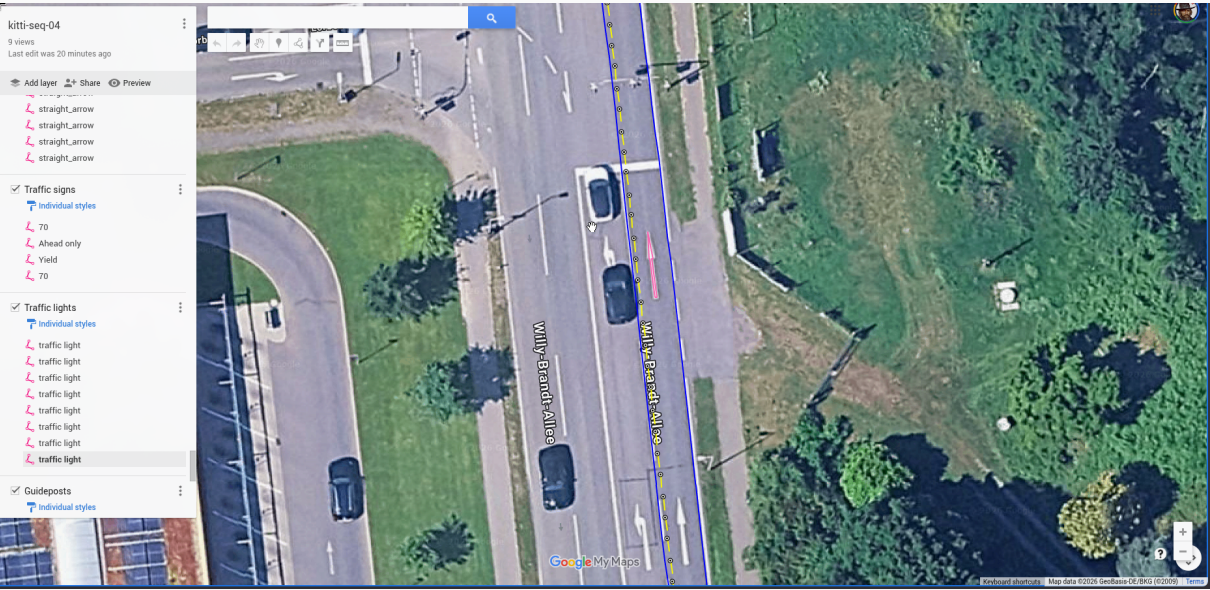

frame_000000.jpg kpts=8071
Building 6800
Car 580
Billboard 276
Person 77
Pole 63
Vegetation 62


frame_000000.jpg kpts=8071
4.6 m
70.9 m

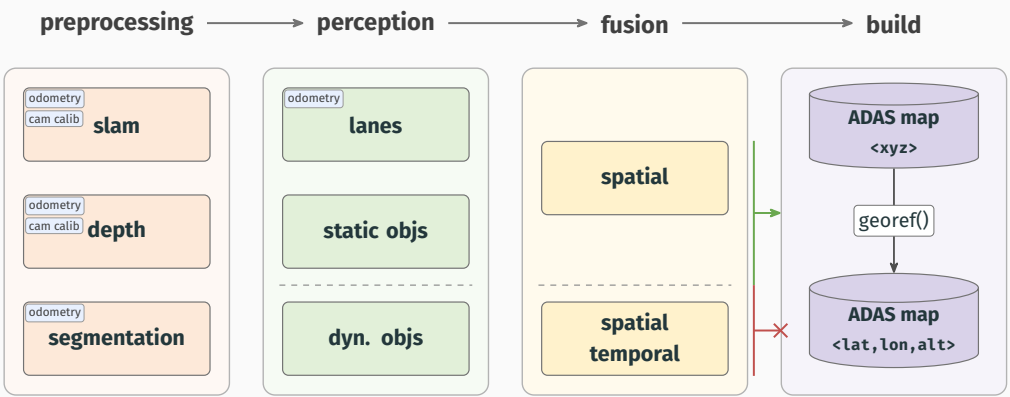

frame 000/270 ego-lane=green border=black | yellow=arrow magenta=sign green=light white=guidepost | red=moving orange=pot_dyn (in-box) (poles excluded)
car_pot_dyn
car_DYN
car_DYN
A0
70
70




frame 235
heading 353.7 deg
49.03661, 8.39447


kitti-seq-04
9 views
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straight_arrow
straight_arrow
straight_arrow
straight_arrow
Traffic signs
Individual styles
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70
Traffic lights
Individual styles
traffic light
traffic light
traffic light
traffic light
traffic light
traffic light
traffic light
traffic light
Guideposts
Individual styles
Willy-Brandt-Allee
Willy-Brandt-Allee
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Map Building Workflow



Questions?